

Re-sit exam August 2025 – Solutions

TFE4120

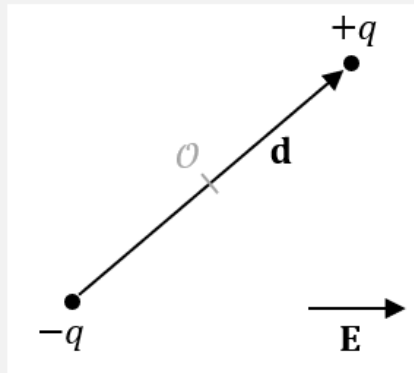
Exercise 1

Exercise 1: 10 marks (2, 6 and 2)

a) A neutral atom with atomic polarizability α is placed in an electric field $\mathbf{E}$. What is the induced dipole moment of the atom?

b) A polar molecule is described as a dipole with dipole moment $\mathbf{p} = q\mathbf{d}$, where $\pm q$ is the charge on either side of the molecule and $\mathbf{d}$ is the separation vector from the negative end of the molecule to the positive, as shown in the figure below. Show that the torque experienced by this molecule due to a uniform electric field $\mathbf{E}$ is $\mathbf{N} = \mathbf{p} \times \mathbf{E}$.

c) What is the effect of this torque on the molecule?



Solution

a) The induced dipole moment is $\mathbf{p} = \alpha\mathbf{E}$.

b) The molecule will experience a force $\mathbf{F}_+ = q\mathbf{E}$ on the positive end and a force $\mathbf{F}_- = -q\mathbf{E}$ on the negative end. These forces cancel, but we nevertheless have a torque $\mathbf{N}$ on the molecule. We have

$$\begin{aligned}\mathbf{N} &= (\mathbf{r}_+ \times \mathbf{F}_+) + (\mathbf{r}_- \times \mathbf{F}_-) = \left(\frac{\mathbf{d}}{2} \times q\mathbf{E}\right) + \left(-\frac{\mathbf{d}}{2} \times -q\mathbf{E}\right) = \frac{q}{2}(\mathbf{d} \times \mathbf{E} + \mathbf{d} \times \mathbf{E}) \\ &= q\mathbf{d} \times \mathbf{E} = \mathbf{p} \times \mathbf{E},\end{aligned}$$

as stated in the task.

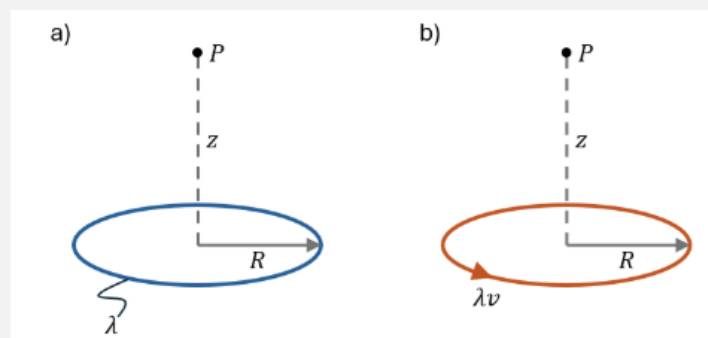
c) The torque $\mathbf{N}$ causes the molecule to line up parallel to $\mathbf{E}$.

Exercise 2

Exercise 2: 20 marks (10 and 10)

a) Find the electric field a distance z above a circular loop of wire carrying a uniform line charge λ . The loop has radius R .

b) The line charge now moves at a constant velocity v around the loop, meaning that the loop carries a steady current. Find the magnetic field a distance z above the loop.



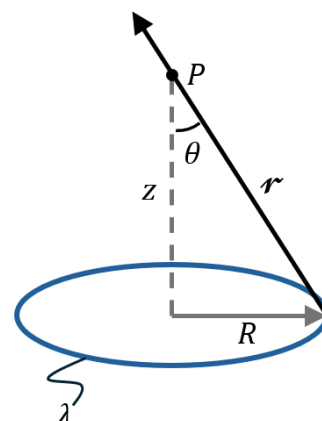
Solution

a) Due to the geometry, only the E_z components survive, while the others cancel. We get

$$E_z = \frac{1}{4\pi\epsilon_0} \int \frac{\lambda dl}{r^2} \cos \theta.$$

We also have that $r^2 = R^2 + z^2$, $\cos \theta = z/r$ and $\int dl = 2\pi R$. Thus, we get that

$$\mathbf{E} = E_z \hat{\mathbf{z}} = \frac{\lambda}{4\pi\epsilon_0} \frac{2\pi R z}{r^3} \hat{\mathbf{z}} = \frac{1}{4\pi\epsilon_0} \frac{\lambda 2\pi R z}{(R^2 + z^2)^{3/2}} \hat{\mathbf{z}}.$$

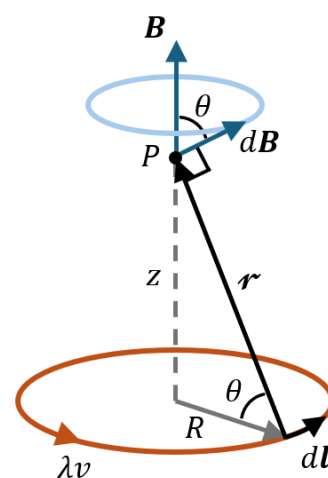


b) The magnetic field attributed to the loop segment dl' is $d\mathbf{B}$. When integrating dl' around the loop, $d\mathbf{B}$ sweeps out a cone. The horizontal components of the field cancel and we are left with $\mathbf{B}$ pointing upward, having only a B_z component. We get

$$B_z = \frac{\mu_0}{4\pi} I \int \frac{dl'}{r^2} \cos \theta.$$

We also have that $r^2 = R^2 + z^2$, $\cos \theta = R/r$ and $\int dl' = 2\pi R$. Thus, we get that

$$\mathbf{B} = B_z \hat{\mathbf{z}} = \frac{\mu_0 I}{4\pi} \frac{2\pi R^2}{r^3} \hat{\mathbf{z}} = \frac{\mu_0 \lambda v}{2} \frac{R^2}{(R^2 + z^2)^{3/2}} \hat{\mathbf{z}}.$$



Exercise 3

Exercise 3: 25 marks (5, 10 and 10)

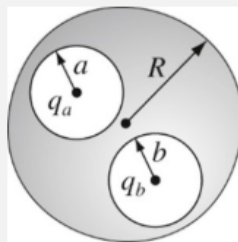
a) Explain how the electric field inside a conductor is always zero. What happens when a conductor is placed in an external electric field?

b) A spherical cavity of radius a is hollowed out from the interior of an uncharged conducting sphere of radius R . A point charge q_a is placed at the center of the cavity.

1. Find the surface charge densities σ_a and σ_R on the surface of the cavity and the surface of the conductor.
2. Find the electric field outside the conductor.
3. Find the electric field within the cavity.
4. What is the force on q_a ?

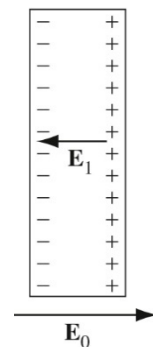
c) A second cavity is now hollowed out from the conductor. This cavity is also spherical and has radius b . At the center of this cavity, a point charge q_b is placed.

1. Find the surface charge densities σ_a , σ_b and σ_R on the surface of each cavity and the surface of the conductor.
2. Find the electric field outside the conductor.
3. Find the electric field within each of the two cavities.
4. What is the force on q_a and q_b ?



Solution

- a) Putting a conductor in an external field causes free charges to move: positive flow along $\mathbf{E}_0$ and negative flow opposite to $\mathbf{E}_0$, see figure. At the boundaries of the material, charges pile up. These charges produce a field of their own, $\mathbf{E}_1$, in the opposite direction to $\mathbf{E}_0$. The charges flow until the induced field cancels the original field, so that the electric field inside the conductor is precisely zero. This happens almost instantaneously.



- b) In this case we have one spherical cavity.

1. $\sigma_a = -\frac{q_a}{4\pi a^2}$ and $\sigma_R = \frac{q_a}{4\pi R^2}$.
2. It is the field of the charge q_a distributed over the outer surface of the conductor. Using Gauss's law with a Gaussian surface consisting of a sphere with radius $r > R$, concentric to the conducting sphere, we find that

$$\oint \mathbf{E} \cdot d\mathbf{a} = E \cdot 4\pi r^2 = \frac{1}{\epsilon_0} q_a \Rightarrow E = \frac{1}{4\pi\epsilon_0} \frac{q_a}{r^2}.$$

Thus, the electric field outside the conductor is $\mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{q_a}{r^2} \hat{\mathbf{r}}$, where $\mathbf{r}$ is the vector from the center of the conducting sphere.

3. Gauss's law with a Gaussian surface consisting of a sphere with radius $r_a < a$, concentric to the spherical cavity, gives that

$$\oint \mathbf{E} \cdot d\mathbf{a} = E \cdot 4\pi r_a^2 = \frac{1}{\epsilon_0} q_a \Rightarrow E = \frac{1}{4\pi\epsilon_0} \frac{q_a}{r_a^2}.$$

Thus, the electric field inside the cavity is $\mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{q_a}{r_a^2} \hat{\mathbf{r}}_a$, where $\mathbf{r}_a$ is the vector from the center of the cavity.

4. Zero.

c) In this case we have two spherical cavities.

1. $\sigma_a = -\frac{q_a}{4\pi a^2}$, $\sigma_b = -\frac{q_b}{4\pi b^2}$ and $\sigma_R = \frac{q_a+q_b}{4\pi R^2}$.
2. It is the field of the charge $q_a + q_b$ distributed over the outer surface of the conductor. Using Gauss's law with a Gaussian surface consisting of a sphere with radius $r > R$, concentric to the conducting sphere, we find that

$$\oint \mathbf{E} \cdot d\mathbf{a} = E \cdot 4\pi r^2 = \frac{1}{\epsilon_0} (q_a + q_b) \Rightarrow E = \frac{1}{4\pi\epsilon_0} \frac{q_a + q_b}{r^2}.$$

Thus, the electric field outside the conductor is $\mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{q_a+q_b}{r^2} \hat{\mathbf{r}}$, where $\mathbf{r}$ is the vector from the center of the conducting sphere.

3. Similarly to task b) c., we find the fields within each cavity to be $\mathbf{E}_a = \frac{1}{4\pi\epsilon_0} \frac{q_a}{r_a^2} \hat{\mathbf{r}}_a$, where $\mathbf{r}_a$ is the vector from the center of the cavity with radius a , and $\mathbf{E}_b = \frac{1}{4\pi\epsilon_0} \frac{q_b}{r_b^2} \hat{\mathbf{r}}_b$, where $\mathbf{r}_b$ is the vector from the center of the cavity with radius b .
4. Zero.

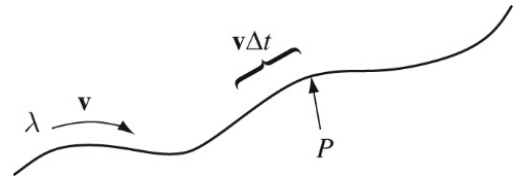
Exercise 4

Exercise 4: 5 marks

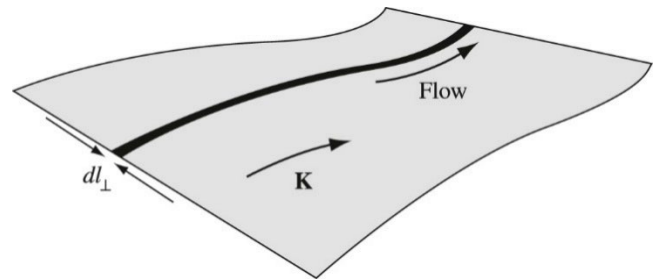
Define the current (one dimension), surface current density (two dimensions) and volume current density (three dimensions) for mobile charges *and* constant flow of charge. Define all quantities you involve, with words and in one sketch per case.

Solution

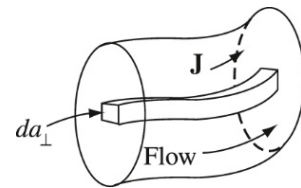
1D: With a mobile line charge λ moving with velocity $\mathbf{v}$, the current is $\mathbf{I} = \lambda \mathbf{v}$.



2D: The surface current density is $\mathbf{K} \equiv \frac{d\mathbf{I}}{dl_{\perp}}$ per unit width. With a mobile surface charge density σ moving with velocity $\mathbf{v}$, we have $\mathbf{K} = \sigma \mathbf{v}$.



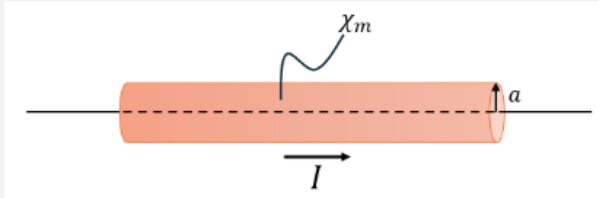
3D: The volume current density is $\mathbf{J} \equiv \frac{d\mathbf{I}}{da_{\perp}}$ per unit area. With a mobile volume charge density ρ moving with velocity $\mathbf{v}$, we have $\mathbf{J} = \rho \mathbf{v}$.



Exercise 5

Exercise 5: 20 marks (2, 8, 8 and 2)

Consider the long straight wire of radius a shown in the figure, made from copper. Copper is a linear material with susceptibility $\chi_m = -9.7 \cdot 10^{-6}$. The wire carries a uniform current I .



- Considering the given susceptibility of copper, is copper diamagnetic or paramagnetic? What does this indicate about the direction of the magnetization $\mathbf{M}$ compared to the direction of the applied field $\mathbf{B}$?
- What is the magnetic field a distance s away from the wire axis? Note that we ask for the field a distance away from the *axis* of the wire and do not specify whether s is greater or smaller than the radius a . In other words, you should find the magnetic field for points both inside and outside the material.
- Find all the bound currents.
- What is the net bound current flowing along the wire?

Solution

- Copper has a susceptibility $\chi_m < 0$, meaning that it is diamagnetic. The magnetization points in the opposite direction to the applied field.
- We use Ampère's law $\oint \mathbf{H} \cdot d\mathbf{l} = I_{fenc}$. The current is longitudinal, meaning that $\mathbf{B}$ and thereby $\mathbf{H}$ are circumferential. We select an Amperian loop in the form of a circle concentric with the wire and integrate. Inside the wire, the current enclosed depends on s , and we have $I_{fenc} = \frac{I}{a^2\pi} \cdot s^2\pi = \frac{Is^2}{a^2}$. Outside the wire, the current enclosed is the total current in the wire, I . Thus, we find

$$\oint \mathbf{H} \cdot d\mathbf{l} = H(2\pi s) = I_{fenc} = \begin{cases} \frac{Is^2}{a^2} & s < a \\ I & s > a \end{cases}$$

This gives

$$H = \begin{cases} \frac{Is}{2\pi a^2} & s < a \\ \frac{I}{2\pi s} & s > a \end{cases} \Rightarrow B = \mu H = \begin{cases} \frac{\mu_0(1 + \chi_m)Is}{2\pi a^2} & s < a \\ \frac{\mu_0 I}{2\pi s} & s > a \end{cases}$$

running circumferential to the wire, with $\chi_m = -9.7 \cdot 10^{-6}$.

- For the volume bound current, $\mathbf{J}_b = \chi_m \mathbf{J}_f$ with $J_f = I/\pi a^2$ gives $J_b = \chi_m I/\pi a^2 = -9.7 \cdot 10^{-6} \cdot I/\pi a^2$ in the same direction as I . For the surface bound current, we

use $\mathbf{K}_b = \mathbf{M} \times \hat{\mathbf{n}}$ and $\mathbf{M} = \chi_m \mathbf{H}$ to find $K_b = \chi_m I / 2\pi a = -9.7 \cdot 10^{-6} \cdot I / 2\pi a$ in the opposite direction to I .

- d) The net bound current along the wire is $I_b = J_b(\pi a^2) + K_b(2\pi a) = \chi_m I - \chi_m I = 0$. This makes sense since we do not expect to have a flow of *bound* current.

Exercise 6

Exercise 6: 20 marks (4, 8 and 8)

a) Provide the definitions of the two types of bound current. What is the physical meaning of bound current and what is their relation to the field of a magnetized object? Draw a parallel to the case with the electric field of a polarized object.

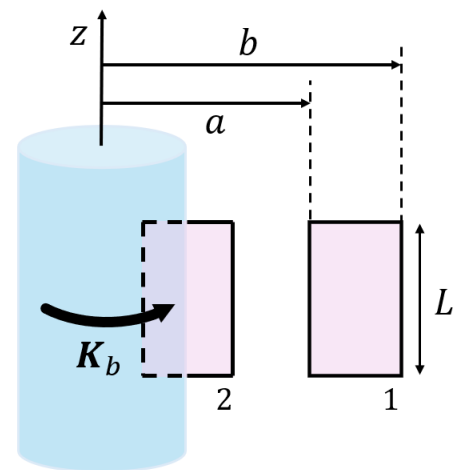
b) An infinitely long cylinder of radius R carries a uniform magnetization $\mathbf{M}$ parallel to its axis. Find the magnetic field due to $\mathbf{M}$ for points inside and outside the cylinder, using the bound currents.

c) Consider again an infinitely long cylinder of radius R , this time carrying a "frozen-in" magnetization $\mathbf{M} = 3s^2\hat{\mathbf{z}}$ parallel to its axis, with s being the distance from the axis. Find the magnetic field due to $\mathbf{M}$ for points inside and outside the cylinder, using the bound currents.

Solution

a) Bound currents are the equivalent to the bound charges we have seen in the electrical case. The volume bound current is $\mathbf{J}_b = \nabla \times \mathbf{M}$ and the surface bound current is $\mathbf{K}_b = \mathbf{M} \times \hat{\mathbf{n}}$, where $\hat{\mathbf{n}}$ is the unit normal vector. The magnetic field of a magnetized object with magnetization $\mathbf{M}$ is the same as the field produced by a bound volume current $\mathbf{J}_b = \nabla \times \mathbf{M}$ throughout the volume of the object and a bound surface current $\mathbf{K}_b = \mathbf{M} \times \hat{\mathbf{n}}$ on its boundary. Here we notice a parallel to the electrical case, where the electric field of a polarized object with polarization $\mathbf{P}$ is the same as the field from a bound volume charge $\rho_b = -\nabla \cdot \mathbf{P}$ in the volume of the object and a bound surface charge $\sigma_b = \mathbf{P} \cdot \hat{\mathbf{n}}$ on its boundary.

b) We have $\mathbf{M} = M\hat{\mathbf{z}}$ meaning that $\mathbf{J}_b = \nabla \times \mathbf{M} = \mathbf{0}$ and $\mathbf{K}_b = \mathbf{M} \times \hat{\mathbf{n}} = M\hat{\boldsymbol{\phi}}$. The magnetic field is that of a surface current $\mathbf{K}_b = M\hat{\boldsymbol{\phi}}$. This is a solenoid, and we find the field using Ampère's law. The direction of the magnetic field due to this current is parallel to the axis of the cylinder, along the z direction. We select two Amperian loops as shown in the figure.



Outside the cylinder (using loop 1):

$$\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 I_{enc} = 0 \Rightarrow (B(a) - B(b))L = 0$$

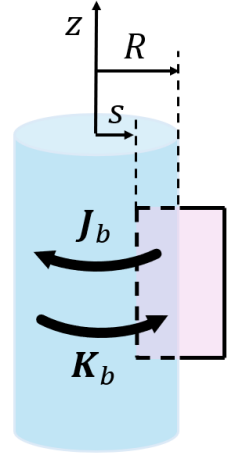
$\Rightarrow B(a) = B(b)$. The magnetic field should go to zero far away from the cylinder, and we now know that it is the same at every point outside the cylinder. Thus, the magnetic field must be zero everywhere outside the cylinder.

Inside the cylinder (using loop 2): $\oint \mathbf{B} \cdot d\mathbf{l} = BL = \mu_0 I_{enc} = \mu_0 K_b L = \mu_0 M L \Rightarrow \mathbf{B} = \mu_0 M \hat{\mathbf{z}}$ inside the cylinder. The side of the loop that is outside the cylinder contributes nothing, since the field is zero there.

The field is thereby found to be

$$\mathbf{B} = \begin{cases} \mu_0 M \hat{\mathbf{z}} = \mu_0 \mathbf{M} & \text{inside the cylinder,} \\ \mathbf{0} & \text{outside the cylinder.} \end{cases}$$

- c) Now we have $\mathbf{M} = 3s^2 \hat{\mathbf{z}}$. The bound currents are $\mathbf{J}_b = \nabla \times \mathbf{M} = -6s \hat{\boldsymbol{\phi}}$ and $\mathbf{K}_b = \mathbf{M} \times \hat{\mathbf{n}} = 3R^2 \hat{\boldsymbol{\phi}}$. The magnetized cylinder we have is essentially a superposition of solenoids, whereby the field outside the cylinder is zero (see task b). Like before, the magnetic field points upwards in the z direction. To find the field inside the cylinder we use Ampère's law with the Amperian loop shown in the figure, with inner side at radius s and outer side outside the cylinder. Now we need to consider both surface and volume bound current. We find



$$\oint \mathbf{B} \cdot d\mathbf{l} = BL = \mu_0 I_{enc} = \mu_0 \left[\int J_b da + K_b L \right]$$

with

$$\int J_b da = L \cdot \int_s^R -6s' ds' = -3L(R^2 - s^2),$$

$$K_b L = 3R^2 L.$$

Thus,

$$BL = \mu_0 L [-3(R^2 - s^2) + 3R^2] = \mu_0 L \cdot 3s^2.$$

We get

$$\mathbf{B} = \begin{cases} \mu_0 3s^2 \hat{\mathbf{z}} & \text{inside the cylinder,} \\ \mathbf{0} & \text{outside the cylinder.} \end{cases}$$